Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

RE Examination

Max. Marks: 100 July 2025 Duration: 3hrs.
Class: T.E. (V) Computer Engineering Course Code: CS304
Name of the Course: Distributed Computing

Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Q. No.		Max Marks	CO-BL
Q. 1 A	Illustrate how distributed computing is applied in real-world systems and identify how common issues arise during implementation.	10	1-3
Q. 1 B	Demonstrate the use of each message communication model with real-life distributed system scenarios.	10	1-3
Q. 2 A	How are message-based communication and stream-based communication different? Give examples of where you might use each one.	10	1-3
Q. 2 B	Explain the significance of the RMI registry in the distributed healthcare application you are developing. Discuss its role in facilitating communication between the client applications, the server-side components handling medical consultations and record access, and any auxiliary services involved.	10	2-3
Q. 3 A	How does a distributed file system like HDFS help manage and store such big data across multiple computers in a distributed system? Explain the key components and how it ensures reliability and scalability.	10	2-2
Q. 3 B	Analyse different types of code migration (strong vs weak, sender-initiated vs receiver-initiated) and compare their advantages and limitations.	10	3-4
Q. 4 A	Why logical clocks are required in distributed systems? Analyse the need for logical clocks and illustrate the working of Lamport's Logical Clock algorithm with a suitable example. **OR** How does the Suzuki-Kasami algorithm help multiple computers take turns safely while using a shared resource in a network? Explain the steps of the algorithm with a simple example.	10	3-3
Q. 4 B	Why do we need a coordinator in a distributed system? Choose one algorithm and explain how it helps select a coordinator, using a simple example.	10	3-3
Q. 5 A	You are making a travel app where a piece of code moves between servers to collect live offers. What is code migration? What problems can happen during this movement? Give two smart ways to solve problems.	10	4-3
Q. 5 B	How does replication help solve challenges like performance, fault tolerance, and availability? Explain any two replication schemes with examples.	10	4-3